

Ground measurements of the polarized bidirectional reflectance of snow in the near-infrared spectral domain: Comparisons with model results

Catherine Leroux,¹ Jean-Luc Deuzé,² Philippe Goloub,² Claude Sergent,³ and Michel Fily¹

Abstract. Ground-based measurements were carried out in the French Alps in March 1995 to study the bidirectional and polarized bidirectional reflectances of various snow covers. Atmospheric (optical thickness) and snow characteristics (grain size and type) were measured simultaneously. The ground reflectance data are compared with snow reflectance modeling results. A radiative transfer model based on the adding-doubling method is used to compute the snow reflectance. The single-scattering parameters are determined by a Mie or a ray-tracing code depending on the particle shape introduced in the model (spherical or hexagonal ice particles). Special attention is paid to the comparisons in the near-infrared spectral range (1.65 μm) because of the sensitivity of snow reflectance to the grain size and shape. In the visible and near-infrared domains of the solar spectrum, the amplitude of the polarized signal is very low and independent of the snow grain size. The agreement between measurements and model results is good if hexagonal particles are introduced in the model instead of the spherical snow grains commonly used in reflectance studies.

1. Introduction

The knowledge of the snow albedo at the Earth's surface and its spatial and temporal variability is fundamental for climatologic studies [Marshall and Oglesby, 1994]. Hemispheric reflectance measurements are needed to determine the snow albedo [Hall et al., 1992], whereas with remote sensing, only the snow bidirectional reflectance is measured. Satellite remote sensing is adequate to deduce snow characteristics from space in remote areas such as polar ice sheets [Bourdelle and Fily, 1993; Steffen et al., 1993]. However, to improve the interpretation of remote sensing data, ground measurements are necessary. By choosing suitable in situ sites with various snow cover characteristics and different solar angles, ground measurements contribute to a better understanding of the relationships between the reflectance and the optical and physical parameters of snow. One of these parameters is the snow grain size which is a relevant variable for snow metamorphism studies and for the snow albedo modeling.

Much literature is available regarding measurements of snow hemispheric reflectance or hemispherical-directional reflectance [Grenfell and Maykut, 1977; Wiscombe and Warren, 1980; Sergent et al., 1993, 1998], but less concerns bidirectional snow reflectance [Steffen, 1987; Grenfell et al., 1994]. The main goal of this paper is to analyze the bidirectional and polarized bidirectional reflectances of snow in relation to the geometry and size of the snow grains and to validate our snow reflectance model. For this purpose we compare ground measurements with model results.

Polarization is a relative newcomer to terrestrial remote sensing in the solar spectrum. There are very few past polarization studies of planetary surfaces [Dollfus, 1961; Hapke, 1993] although polarization has been used extensively in astronomy for characterization of planetary atmospheres [Hansen and Hovenier, 1974; Colin, 1980; Hovenier and Van der Mee, 1983]. The polarization of the reflected signal is now analyzed because it could provide additional information for snow cover characterization, and it is a useful tool for snow/cloud discrimination [Goloub et al., 1992]. It may also help to interpret data acquired by the POLDER (POLarization and Directionality of Earth's Reflectance) instrument [Deschamps et al., 1994] which was launched in August 1996 onboard the ADEOS (Advanced Earth Observing Satellite) Japanese platform. This sensor measured the state of polarization of the light (radiance, polarized radiance, and direction of the plane of polarization) in three out of its eight visible and near-infrared spectral band channels. These data are obtained by a combination of measurements through three analyzer positions. POLDER also observed the same target from 13 directions during a single satellite pass. Our study can also serve to interpret data from the ERS/ATSR (Along Track Scanning Radiometer) and the future SPOT4/vegetation radiometers which both have a channel centered on the 1.6 μm wavelength.

After a description of the field instruments and the measurement procedure (section 2), we describe the snow reflectance model and discuss the influence of the atmosphere on the incident radiation incoming above the surface in section 3. In the last section we compare between measurements and model results for both the bidirectional reflectance and the polarized bidirectional reflectance.

2. Ground Measurements

2.1. Reflectance Measurements

A photopolarimeter was developed by the "Laboratoire d'Optique Atmosphérique" in Lille, France, to measure the bidirectional polarized radiance of the atmosphere and ground surfaces in the field (see definitions in section 3.4). The filters corresponding to the four wavelengths of this instrument are

¹Laboratoire de Glaciologie et Géophysique de l'Environnement, Centre National de la Recherche Scientifique, Université Joseph Fourier, Saint Martin d'Hères, France

²Laboratoire d'Optique Atmosphérique, Université des Sciences et Technologies de Lille, Villeneuve d'Ascq, France

³Centre d'Etudes de la Neige, Météo France, Saint Martin d'Hères, France

centered at 0.45, 0.65, 0.85, and 1.65 μm with bandwidths of 0.1 μm for the near-infrared and 0.04 μm for the other channels. Radiances and polarized radiances are obtained by combining measurements made at three directions of the analyzer. The instrument field of view (FOV) is around 6°, which corresponds to a ground nadir pixel diameter of about 0.14 m when the instrument is mounted 2 m above the surface. Measurements were carried out in the principal plane and in the perpendicular plane. The principal plane offers the greatest variability of scattering angles. Scanning of the reflectance measurements is done automatically in the principal plane for view angles (θ') from -80° to 80°. By convention, θ' is positive when the observation is made in the vertical half plane, facing the Sun. The data are stored in a PC. The digital counts are converted to radiance using calibration coefficients determined in the laboratory using an integrating sphere. The radiances are then normalized to the solar irradiance at the top of the atmosphere. Bréon *et al.* [1995] report that this quantity (which is a reflectance) is well adapted to polarization studies because the diffuse part of the downwelling irradiance generates negligible polarization after reflection at the surface.

The snow reflectance was measured over three different sites in the French Alps in March 1995. We selected four data sets (Table

1) acquired under clear sky conditions to avoid unpredictable changes in the incident solar flux during measurements (Figure 1).

For set 1, acquired in the plane perpendicular to the principal plane, the scattering angle Θ , between light rays before and after scattering, is determined by the following equation:

$$\cos(\pi - \Theta) = \cos\theta_0 \cos\theta' - \sin\theta_0 \sin\theta' \cos(\Delta\phi) \quad (1)$$

where θ_0 is the solar zenith angle, and $\Delta\phi$ is the difference between the solar azimuthal angle (ϕ_0) and the view azimuthal angle (ϕ') ($|\Delta\phi| = 90^\circ$ for measurements in the perpendicular plane). The range of scattering angles is more limited in the perpendicular plane than in the principal plane.

In the visible part of the solar spectrum, the reflectance depends mainly on contamination by dust and soot and is relatively insensitive to snow grain size [Warren, 1982; Dozier, 1989; Fily *et al.*, 1998]. The opposite trends are observed in the near-infrared spectral range. At 0.85 μm , there is still a nonnegligible effect of soot on the reflectance [Grenfell *et al.*, 1994]. This paper is focused on the influence of the snow grain type and size on the bidirectional and polarized bidirectional reflectances; we therefore present only results in the 1.65 μm wavelength channel. Another advantage of

Table 1. Ground Site Characteristics and Data Sets Selected for Comparisons With Model Simulations

Set Number	1	2	3	4
Date	March 11	March 11	March 12	March 13
Location	Col de Porte	Col de Porte	Col du Lautaret	Chamrousse
Elevation, m	1300	1300	2020	2250
Measurements plane	perpendicular	principal	principal	principal
θ_0 (°)	63.47	62.36	55.74	48.23
Θ (°)	94 to 116	38 to 180	45 to 180	52 to 180
τ aerosols	0.049	0.049	0.021	0.015
Type of snow (surface layer)	clustered rounded grains	clustered rounded grains	faceted and fine particles	faceted crystals and surface hoar
rcvx, μm (surface layer)	263	263	134	141
Density, kg/m^3 (surface layer)	470	470	no measurement	no measurement
Type of snow (below the surface)	fine grains	fine grains	clustered rounded grains	clustered rounded grains
rcvx, μm (below the surface)	120	120	288	164
Density, kg/m^3 (below the surface)	360	360	470	430

Value θ_0 is the solar zenith angle and Θ the scattering angle. The measurements are made parallel or perpendicular to the principal plane. Aerosol optical thickness (τ aerosols) is given at 0.55 μm wavelength and deduced from the sun-photometer measurements. rcvx is the mean convex radius.

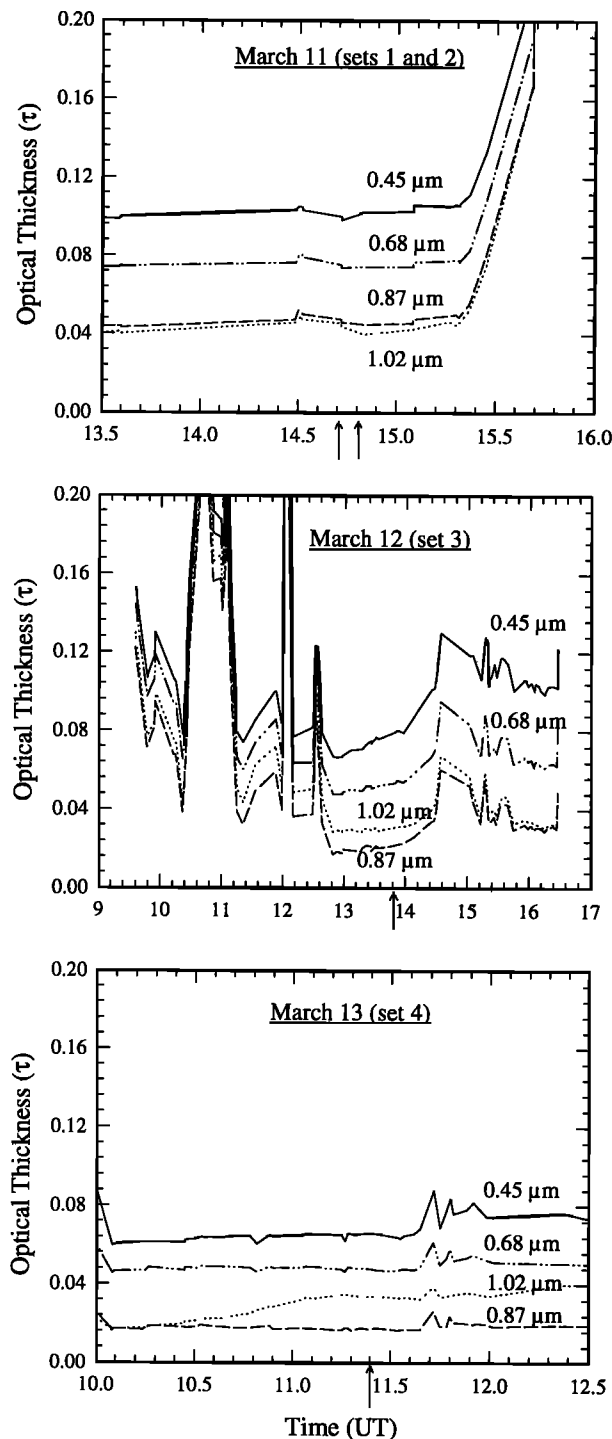


Figure 1. Atmospheric optical thickness acquired by the Sunphotometer during the measurement campaign. The arrows show the selected data sets (see Table 1).

data acquisition in this spectral band is that the atmospheric effects are smaller than in the visible domain.

2.2. Atmospheric Measurements

Simultaneously a Sunphotometer measured the total (aerosol and molecular) atmospheric optical thickness (τ) from attenuation of the direct Sun irradiance at 0.45, 0.68, 0.87, and 1.02 μm wavelengths. In Figure 1 the atmospheric optical thickness is plotted versus time for each of the three investigated sites. After subtraction of the

molecular contribution, the aerosol optical thicknesses are deduced and interpolated in each channel of the photopolarimeter using Angström's law:

$$\tau(\lambda') = \tau(\lambda) \left(\frac{\lambda'}{\lambda} \right)^{-\alpha}, \quad (2)$$

α is first computed from measurements in the 0.45 and 0.87 μm Sunphotometer channels. Then $\tau(\lambda')$ can be computed at any λ' if $\tau(\lambda)$ is known. The λ' is a photopolarimeter wavelength, and λ is a Sunphotometer channel. Table 1 gives the aerosol optical thickness for the three measurement sites at the 0.55 μm wavelength. The aerosol optical thickness decreases strongly as elevation increases. The atmospheric conditions associated with the four sets of measurements are well adapted to this study (low optical thickness).

2.3. Snow Characteristics

For each site, the stratigraphy of the snow cover was determined. Snow samples were collected in the field from homogeneous layers and kept in isooctane below 0°C to stop metamorphism [Brun and Pahaut, 1991]. Each sample was analyzed by an image analysis system which computes the mean radius of curvature of randomly chosen crystals [Lesaffre et al., 1998]. The curvature is computed for each pixel of the grain perimeter. Typically, one sample contains 60 to 80 crystals. The mean convex radius is considered to be an objective (independent of the experimenter) size factor to characterize the snow grains [Sergent et al., 1993, 1998]. It will be noted "rcvx" (Table 1).

Though the entire snow cover stratigraphy was determined, we will consider in the model only the first layer of snow with a single snow grain size (rcvx) because in the near-infrared 1.65 μm wavelength, ice absorption is much larger than in the visible domain, and the penetration depth of light is therefore smaller [Warren, 1984].

Figure 2 shows the three snow types present on the surface during the three days of measurements. On the March 11, for data sets 1 and 2, the 2 cm thick surface snow layer was wet with clustered rounded grains. On the March 12 (data set 3) the surface snow layer was 0.5 cm thick and made of a mixture of small faceted crystals and rounded grains, lying over a melt-freeze crust layer. On the March 13 the surface snow layer was 0.5 cm thick with surface hoar lying over a melt-freeze crust.

3. Snow Reflectance Modeling

In this section, we present the snow reflectance model we have developed to account for the polarized signature of the light reflected by the snow surface. First, we review the general features of the polarization formalism, before detailing the snow and atmospheric models separately. Finally, the atmosphere and snow media are combined. All the variables are wavelength dependent.

3.1. State of Polarization

The snow reflectance model we used takes into account the polarized information necessary for comparisons with the measurements of the photopolarimeter. Detailed descriptions of the scattering of light typically express the light as a vector $\vec{L} = (L, Q, U, V)$ of the Stokes parameters, where L , Q , U , and V have been defined by Van de Hulst [1981], Chandrasekhar [1960], and Hovenier and Van der Mee [1983]. Usually, the parameter L represents the intensity, omitting a constant factor common to all four parameters, but in this paper L will be the radiance. Q , U , and V describe the state of polarization. When the

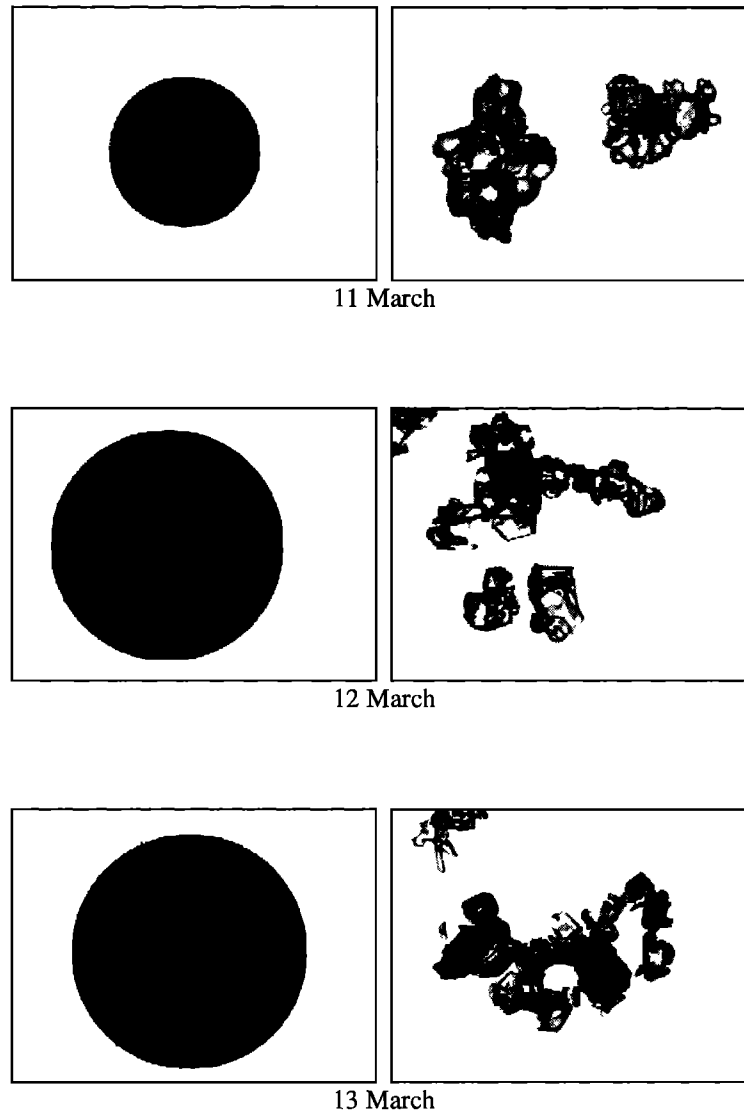


Figure 2. Images of the surface snow grains for the three days of measurements. The scale is given by the circles on the left (3 mm diameter).

light is linearly polarized, the ellipticity V is null. The Stokes parameters are defined with respect to a reference plane. For our calculations we chose the meridian plane which contains the viewing direction. Assuming a linear polarization, we introduce the polarized radiance

$$L_{\text{pol}} = \sqrt{Q^2 + U^2}, \quad (3)$$

which is independent of the reference plane, and the angle χ , $\tan 2\chi = (U/Q)$, which indicates the direction of polarization. Nevertheless, as the polarized signal is mainly generated by primary interactions (reflection, scattering), the polarized vibration is perpendicular or parallel to the scattering plane (containing the Sun and the viewing direction). By convention, L_{pol} is positive in the first case (perpendicular) and negative in the second case (parallel).

3.2. Snow Reflectance Without the Atmospheric Effect

Before introducing the atmospheric effect, the snow reflectance of the surface itself has to be known. The description of the snow reflectance model we developed is detailed in a previous paper

[Leroux *et al.*, 1998]. This model is based on the radiative transfer theory with the use of the adding-doubling method to describe the multiple scattering that occurs in the snowpack [De Haan *et al.*, 1987]. Taking into account the state of polarization, the modeled snow bidirectional reflectance $\tilde{R}_{\text{bn}}$ is expressed by a 4×4 matrix.

To study the influence of the snow grain shape on the reflectance, the snow grains are assumed to be either spherical or hexagonal particles randomly oriented in space. The first geometry has been largely used in previous snow reflectance models [Wiscombe and Warren, 1980; Bohren, 1987; Grenfell, 1992; Fily *et al.*, 1997] and is well adapted to hemispherical snow reflectance modeling. The hexagonal shape is used in cirrus optical property studies [Wendling *et al.* 1979; Coleman and Liou, 1981; Cai and Liou, 1982; Takano and Jayawera, 1985; Brogniez, 1988; Iaquinata *et al.*, 1994; Tanaka and Liou, 1995], and it is also the initial shape of ice crystal germs. The single-scattering parameters and the phase

matrix (input data of the adding-doubling method) are determined by a Mie-scattering code if snow particles are assumed to be spherical or a ray-tracing code for hexagonal crystals [Brogniez, 1988].

In the near-infrared range of the solar spectrum, at 1.65 μm , the snow reflectance depends mainly on the snow grain size and shape. Because the ice absorption at this wavelength is much larger than in the visible domain, the larger the crystal size the smaller the reflectance. This is due to the fact that larger grains are both more absorptive and more forward scattering. In addition, for large wavelengths the scattering in the snow cover becomes closer and closer to single-scattering behavior. In previous studies we have shown that the bidirectional reflectance at 1.65 μm was strongly linked to the phase matrix which depends on the grain shape [Leroux et al., 1998].

For each data set we introduce hexagonal crystals in the model with the same surface/volume ratio as the equivalent sphere with a radius given by the mean convex radius (Table 1). We denote "2c" the diameter of the hexagonal base and "L" the length of the column. Because there is an infinity of plates and column crystals with the same surface/volume ratio as the equivalent sphere, we have arbitrarily chosen particles with a c/L ratio equal to 0.01 for columns and 100 for plates as well as particles with 2c/L=1.

3.3. Atmospheric Downwelling Irradiances

Atmospheric conditions affect snow reflectance measurements [Bréon et al., 1995] and must therefore be taken into account in snow reflectance modeling. The snow cover receives the direct sunlight (F_{dir}) and the diffuse radiation from the sky (F_{diff}). The direct incident irradiance depends on the solar elevation and the attenuation of the light through the atmosphere. The incident natural light is unpolarized and the corresponding Stokes vector is $\tilde{F}_{\text{dir}} = (F_{\text{dir}}, 0, 0, 0)$, where F_{dir} is given by the following equation:

$$F_{\text{dir}} = \mu_0 E_0 \exp(-\tau/\mu_0), \quad (4)$$

μ_0 is the cosine of the solar angle. Value τ , the total atmospheric optical thickness, is the sum of the aerosol optical thickness (see section 2.2) and the molecular optical thickness deduced from an atmospheric model called "6S" (second simulation of the satellite signal in the solar spectrum [Vermote et al., 1997]). At 1.65 μm this model predicts a very low absorption by H_2O and CO_2 gas which is neglected.

The diffuse downwelling irradiance is generated by scattering of the incident light by molecules (Rayleigh-scattering), aerosol (Mie-scattering), and multiple target-environment interactions. These interactions between unpolarized incident light and particles in the sky induce a polarized component in the downwelling diffuse irradiance. The diffuse irradiance was estimated by a successive orders (SO) of scattering code [Deuzé et al., 1989]. In the 1.65 μm channel, diffuse incident irradiance is very low, and we will neglect the polarization of the diffuse irradiance and consider only the first term of the diffuse downwelling Stokes vector. To determine F_{diff} , the aerosols are assumed to be spherical and their size distribution given by a Junge law. The snow surface is assumed to be Lambertian when accounting for the environmental contribution, and in this case its reflectance is the hemispheric reflectance computed by the adding-doubling model with snow grains assumed to be spherical with a radius equal to rcvx (Table 1). If $L_{\text{sky}}^{\downarrow}$ is the downwelling sky radiance, then

$$F_{\text{diff}} = \int_0^{2\pi} \int_0^1 \mu L_{\text{sky}}^{\downarrow}(\mu, \phi) d\mu d\phi. \quad (5)$$

Table 2 summarizes the downwelling atmospheric irradiances for each of the four sets of measurements. The SO model results are compared to simulations without the presence of the atmosphere above the snow surface. It appears that the simulations with the atmosphere are very close to those without the atmosphere, and the snow reflectance is expected to be nearly independent of the diffuse downwelling irradiance at 1.65 μm .

3.4. Snow Reflectance

When the atmosphere is introduced in the model, the radiance reflected by the snow cover is expressed versus the atmospheric downwelling irradiances

$$\tilde{L}(\mu_0, \mu', \phi_0, \phi') = \frac{1}{\pi} [\tilde{R}_{\text{bn}}(\mu_0, \mu', \phi_0, \phi') \tilde{F}_{\text{dir}} + \int_0^{2\pi} \int_0^1 \tilde{R}_{\text{bn}}(\mu, \mu', \phi, \phi') \tilde{L}_{\text{sky}}^{\downarrow}(\mu, \phi) d\mu d\phi]. \quad (6)$$

(μ_0, ϕ_0) is the direction of the incoming solar radiation, and (μ', ϕ') is the viewing direction.

In the near-infrared spectral range the diffuse incident radiation is not isotropic but has a preferential orientation along the incident direction of the Sun. Because the diffuse downwelling irradiance is small (Table 2), we neglected it in equation (6), leading to

$$\tilde{L}(\mu_0, \mu', \phi_0, \phi') = \frac{1}{\pi} [\tilde{R}_{\text{bn}}(\mu_0, \mu', \phi_0, \phi') \tilde{F}_{\text{dir}}]. \quad (7)$$

In order to compare the model results with the measurements (see section 2.1), we compute the following snow BRDF (bidirectional reflectance distribution Function):

$$R(\mu_0, \mu', \phi_0, \phi') = \frac{\pi L(\mu', \phi')}{E_0 \mu_0}, \quad (8)$$

where L is the first element of the radiance Stokes vector.

We also introduce the normalized polarized radiance and the polarized bidirectional reflectance is expressed by

$$R_{\text{pol}}(\mu_0, \mu', \phi_0, \phi') = \frac{\pi \sqrt{Q^2 + U^2}}{E_0 \mu_0}. \quad (9)$$

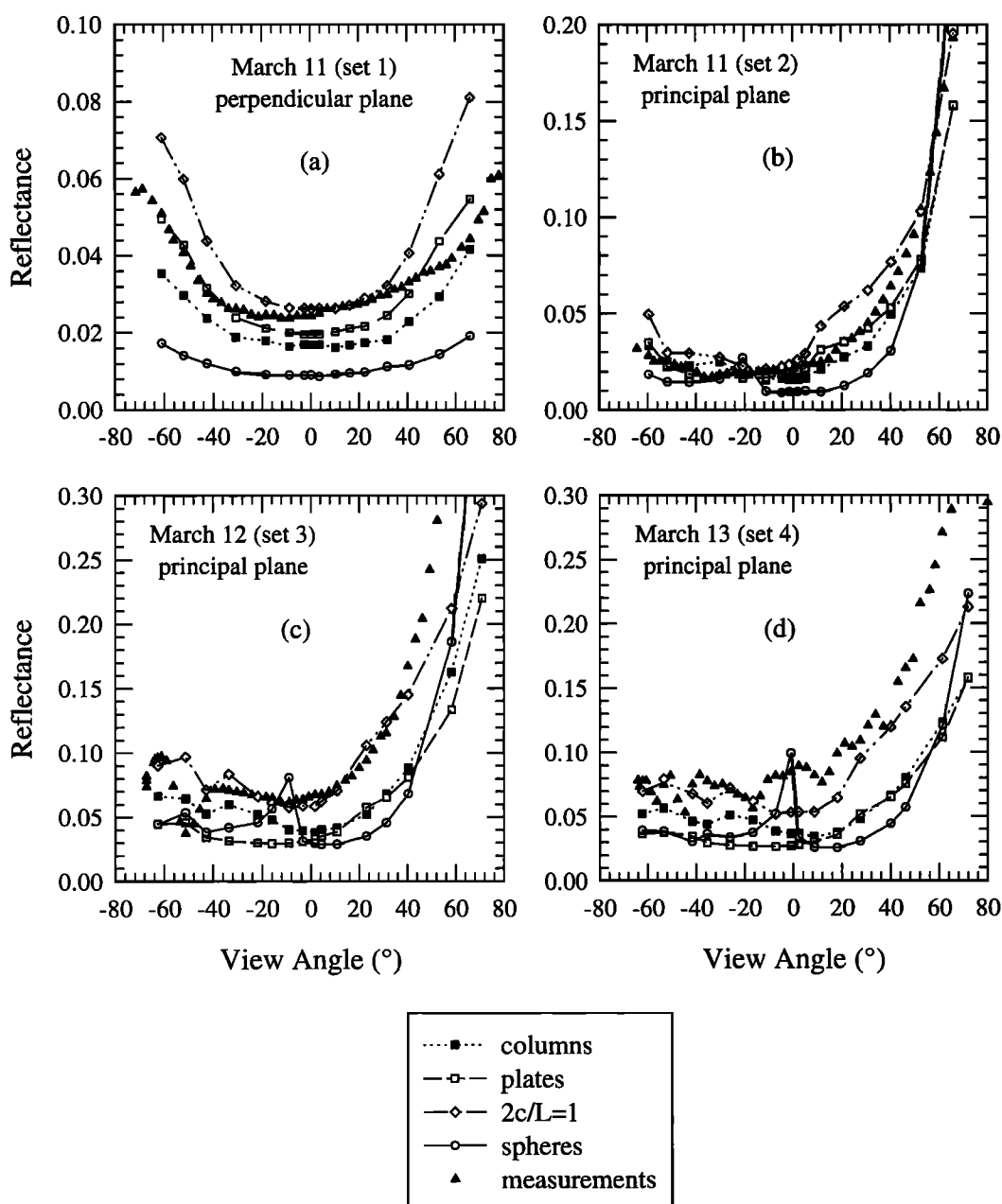
The sign of R_{pol} is a consequence of the convention we have adopted (same sign as L_{pol} ; see section 3.1). Furthermore, we consider that the ellipticity V of the radiance is negligible compared to $\sqrt{Q^2 + U^2}$. Note that $U=0$ in the principal plane.

4. Comparisons Between Measurements and Model Results

In this section, comparisons between the in situ data acquired by the photopolarimeter and the model simulations are presented at 1.65 μm in the principal plane and in the perpendicular plane. We will first focus on the effect of the snow grain shape on the reflectance model, then on the reflectance behavior versus snow grain size. Both bidirectional and polarized reflectances are analyzed. For each data set, all the particles introduced in the model

Table 2. Incident Direct (F_{dir}) and Diffuse (F_{diff}) Irradiances Simulated by the "SO" Model at 1.65 μm With and Without Atmosphere Above Snow Surface

Atmosphere	Set Number							
	1		2		3		4	
	Yes	No	Yes	No	Yes	No	Yes	No
F_{dir}	1.33	1.40	1.38	1.46	1.75	1.77	2.07	2.09
F_{diff}	0.059	0.000	0.060	0.000	0.017	0.000	0.015	0.000

**Figure 3.** Snow bidirectional reflectance (equation (8)) versus the viewing angle at 1.65 μm . Comparisons between measurements and adding-doubling simulations with hexagonal and spherical particles (shape effect).

have the same surface/volume ratio as the equivalent sphere characterized by the mean convex radius determined from snow sample analysis (Table 1). We will not rely on measurements when zenith view angles are less than -60° or greater than 60° because of the environmental effects from surrounding slopes.

4.1. Bidirectional Snow Reflectance

4.1.1. Snow grain shape effect. In the near-infrared range, because of the strong absorption of ice, snow reflectance is sensitive to the snow grain shape. In a previous study [Leroux *et al.*, 1998], we quantified theoretically the snow grain shape effect on the modeled reflectance. Our purpose here is to check if the model is able to reproduce observed data. In Figure 3 the snow bidirectional reflectance is plotted versus the view zenith angle for the four data sets described in Table 1.

In the perpendicular plane (Figure 3a) the symmetry predicted by equation (1) between negative and positive view zenith angles is observed for the computed reflectances. The symmetry is less obvious for the measured reflectance, possibly the result of several effects such as the spatial nonhomogeneity of the target, a possible ground slope or the presence of the forest around the measurement site which can influence the incoming radiation. The general shape of the theoretical curves for spheres and hexagonal crystals agrees well with that of the measurement curve, but only the modeled reflectances from plates and $2c/L=1$ particles are close to the measured reflectances.

In the principal plane the enhanced forward scattering peak is clearly visible for measurements made at large view angles equivalent to small scattering angles (Figures 3b–3d). These curves also present a minimum value due to the detector shadow, located

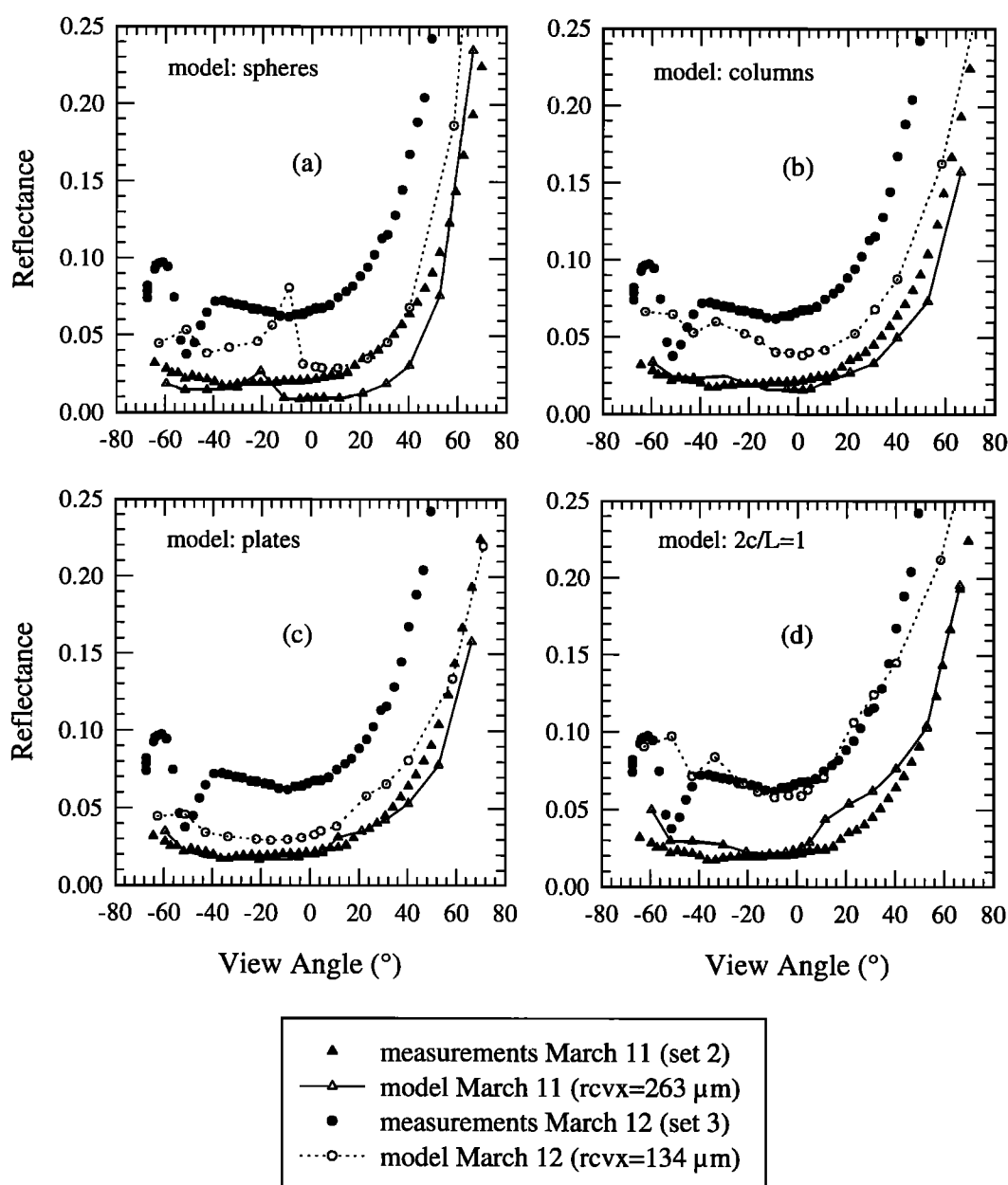


Figure 4. Snow bidirectional reflectance (equation (8)) versus the viewing angle at $1.65 \mu\text{m}$. Comparisons between measurements and adding-doubling simulations with hexagonal and spherical particles (grain size effect).

near -62.4° for data set 2, -55.7° for data set 3, and -48.2° for data set 4 (refer to Table 1 for solar zenith angles values). At $1.65\ \mu\text{m}$ wavelength, snow reflectance is very sensitive to the phase function shape. The modeled reflectance using spheres therefore exhibits the rainbow peak associated with scattering angles near 135° , but this peak is not observed in the measurements. Furthermore, the amplitude of the reflectance simulated with spheres does not agree with the measurements. To fit the measurement curves, it is necessary to introduce different snow grain radii in the model depending on view angles (for instance, in Figure 3b, a snow grain radius of $150\ \mu\text{m}$ is required for small view angles and a $224\ \mu\text{m}$ radius for backscattering). Even when snow grains observed at the surface of the snow cover were rounded (March 11, Table 1), spheres proved to be unsuitable to model the bidirectional reflectance. On the other hand, hexagonal particles reproduce well

the measurements without any adjustment. Except for sets 1 and 2 measured on March 11, where both particles with dimension $2c/L=1$ and plates are able to reproduce the anisotropy of the snow, the crystals with a ratio of $2c/L=1$ provide the best agreement with measurements. The difference between model results based on this crystal shape and measurements is about 20%. This is a mean value averaged over all viewing angles because the discrepancy between the model results and the measurements is not constant with the viewing direction.

4.1.2. Snow grain size effect. In the near-infrared, because of the strong absorption bands, the snow reflectance is sensitive to snow grain size variations. Snow reflectance decreases as absorption increases. We selected data sets 2 and 3 because they provide the largest grain size variability. Figure 4 displays the snow reflectance for the four modeled particle shapes. On each plot we

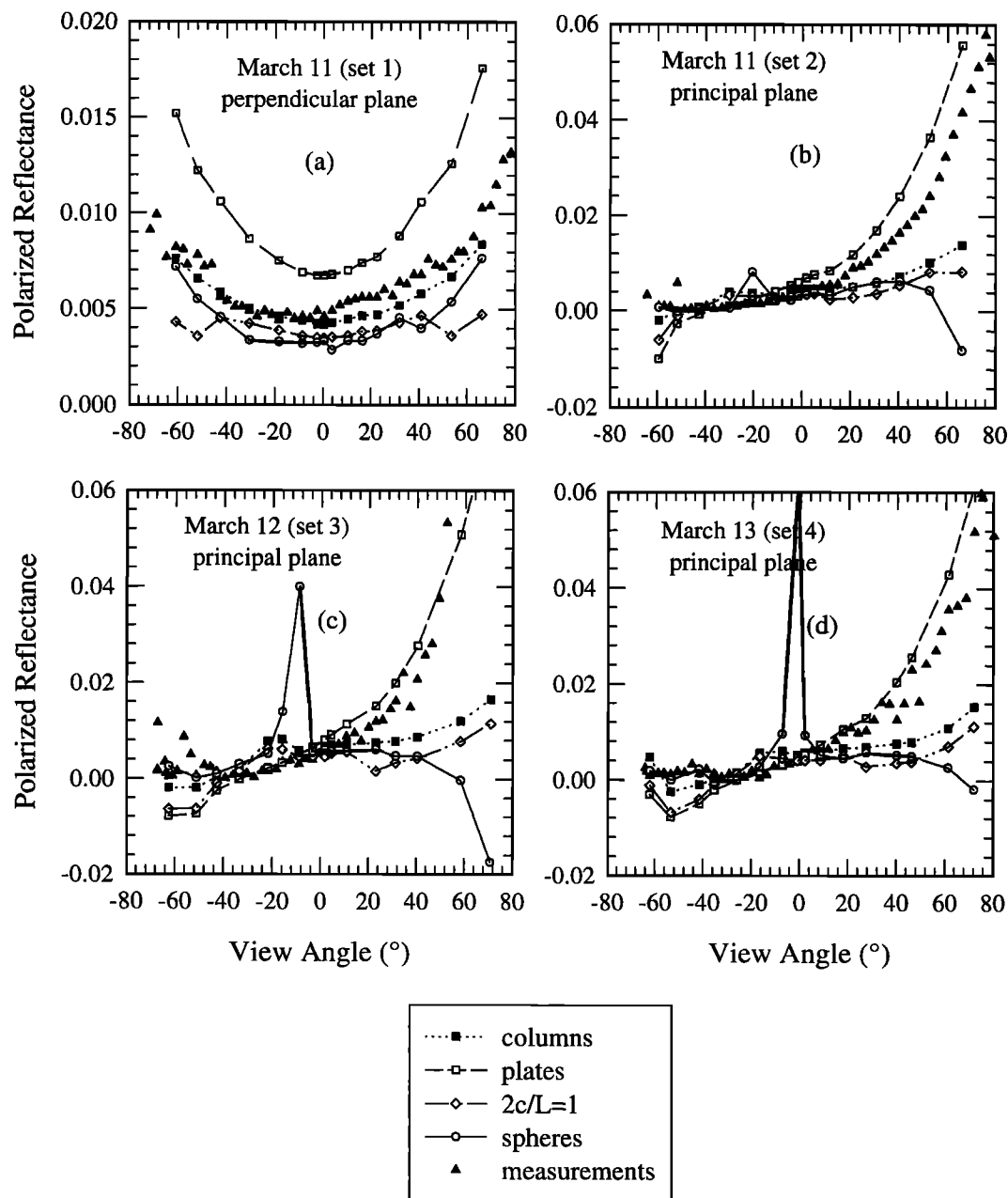


Figure 5. Snow polarized bidirectional reflectance (equation (9)) versus the viewing angle at $1.65\ \mu\text{m}$. Comparisons between measurements and adding-doubling simulations with hexagonal and spherical particles (shape effect).

compared simulations for the two measured data sets. Columns and plates clearly reproduce well the general shape of the measurement curves for data set 2; however, only reflectances modeled from hexagonal crystals such as $2c/L=1$ (Figure 4d) are sensitive to a snow grain variations. Columns and plates are less dependent on snow grain size. The discrepancy between the two theoretical reflectance curves deduced from spherical snow grains (Figure 4a) is maximum in the rainbow region. Spheres for all the data sets and columns and plates for data sets 2 and 3, as already noted, do not match the snow reflectance behavior with respect to the view angle, and furthermore these particle shapes do not account for snow grain size variation.

4.2. Polarized Bidirectional Snow Reflectance

4.2.1. Snow grain shape effect. Figure 5 is equivalent to Figure 3 but for polarized reflectance. The amplitude of the

polarized reflected signal is about 5 times smaller than for the reflectance. The snow polarization reflectance increases in the forward scattering direction just like the reflectance. Polarization concerns only the few first centimeters in the snow cover, explaining why it is also sensitive to snow grain shape. As observed for reflectance, modeling based on spheres does not reproduce the polarization behavior with respect to view angles. The rainbow peak is also produced, and there is even a negative polarization for small scattering angles, which is in opposition with the observations. Except for measurements made in the perpendicular plane, where plates overestimate the polarized reflectance compared with measurements, these particles give the best agreement with the measurements. A difference of about 20% (averaged over viewing angles) is observed for the three data sets in the principal plane. The plates are the only hexagonal crystals which polarize efficiently the reflected radiation for small scattering angles (large viewing angles).

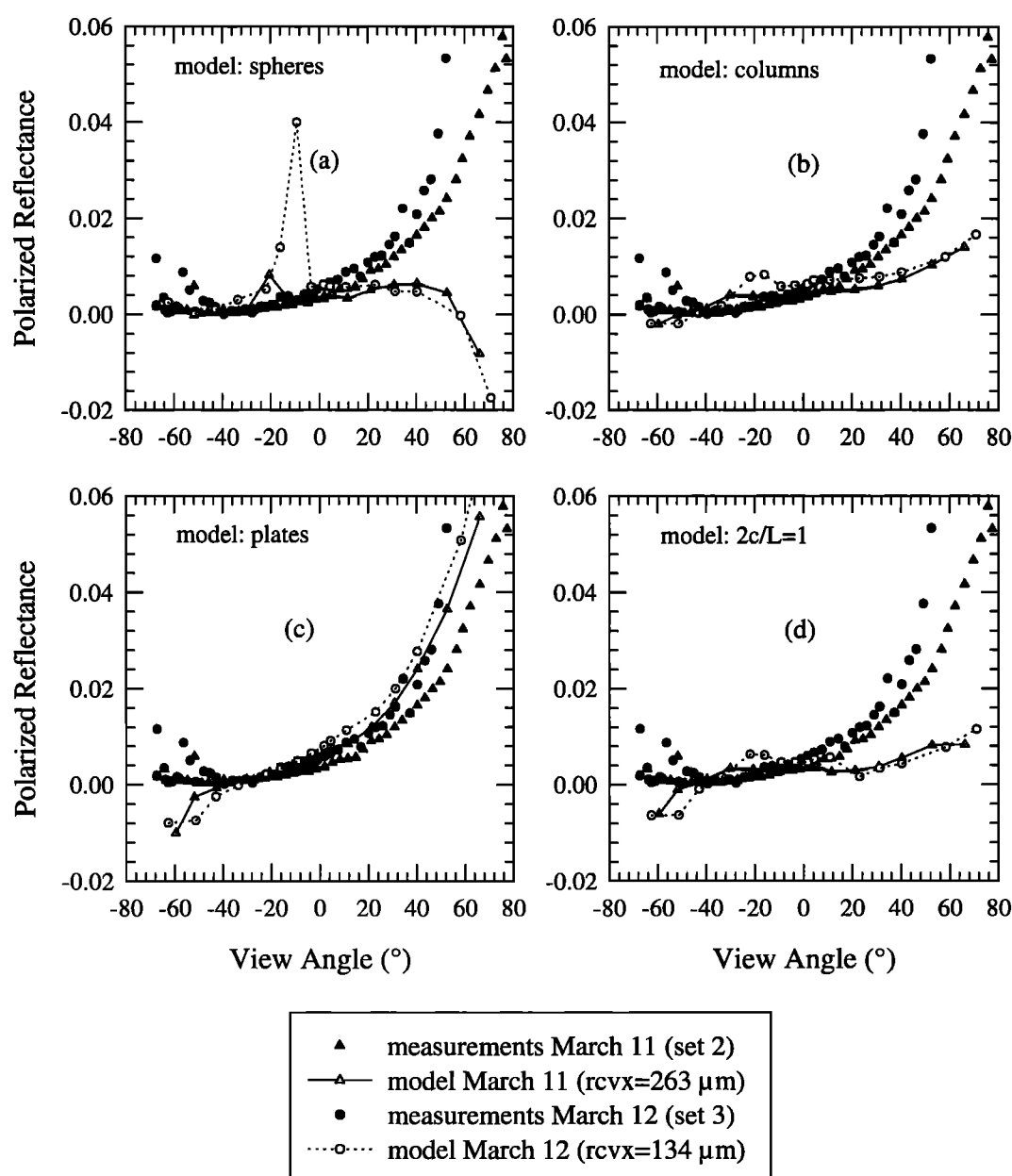


Figure 6. Snow polarized bidirectional reflectance (equation (9)) viewing angle at 1.65 μm. Comparisons between measurements and adding-doubling simulations with hexagonal and spherical particles (grain size effect).

4.2.2. Snow grain size effect. Looking at Figure 6, as opposed to the reflectance behavior, the polarized reflectance measurements are quite invariant with respect to snow grain size. The model using spheres (Figure 6a) shows a large difference between the two simulations in the rainbow region, and the general shape of these two curves moves away from the two observation curves for large viewing angles. Polarization modeled from hexagonal particles is insensitive to snow grain size, but only simulations with plates match the measured reflectances.

4.3. Error Source

Several factors can explain the discrepancies between model and ground data. Firstly, it is important to note the amplitude of the uncertainty of the reflectance measurements which in relative terms is around more or less 5 to 10%.

One cause of differences between model results and observed values could be the topographic effects. The larger the solar zenith angle is, the more the topographic influence prevails. The three ground targets were chosen to be horizontal, but no precise measurements were done. A small slope of 3.5° could induce a modification of 13% in relation to the direct reflectance component.

Another source of uncertainty not accounted for in the model is the bandwidth of the $1.65\ \mu\text{m}$ channel. We assume that the measurements are made at one wavelength. This may be appropriate for the atmosphere but not for the snow reflectance which depends strongly on the wavelength in this spectral region. The integration over the bandwidth could change the results from less than 1% in relative values to 10% depending on the viewing angles.

Concerning data set 3, small oscillations in the measured signal are visible and could be the consequence of the roughness of the surface (footprints). The model does not take into account the effect of the roughness on the BRDF.

A possible orientation of the grains at the surface could explain differences between measurements and model results, but this effect is very difficult to take into account (uneasy to measure the grain orientation in the field and to introduce it in the model). Usually, preferred orientations are only observed for surface hoar.

We only consider in the model the influence of the surface layer. However, in Table 1 characteristics of the snow below the surface layer are also given.

Other effects such as the atmospheric influence are negligible at $1.65\ \mu\text{m}$.

5. Conclusion

Our objectives were first to measure in situ the bidirectional and polarized bidirectional reflectances of three different snow covers in the near-infrared range, then to compare the observations with an analytical snow reflectance model for validation purposes, and finally, to analyze the effect of snow grain size and shape on these reflectances. In the future, more snow grain types should be investigated because our study did not concern all the natural snows which can be encountered in nature. This work should help in interpreting terrestrial bidirectional and polarized data obtained from satellite remote sensing.

Snow bidirectional curves obtained from ground-based measurements in the solar plane exhibit, as predicted by the model, the strong forward scattering related to the snow reflectance anisotropy. We have shown that model simulations based on the spherical representation of snow grains were not able to reproduce either the observed bidirectional reflectances or the observed polarized bidirectional reflectances for all the view angles even if an

adjustment between the model and the measurements is investigated at one wavelength. Model results using hexagonal crystals with dimensions $2c/L=1$ agree well with the bidirectional reflectance measurements and reproduce the sensitivity of the reflectance to snow grain size variations. Plates are suitable to model the measured reflected polarized radiation which is independent of snow grain variations. For both reflectance and polarization studies, no tuning was done to fit the model results and the measurements.

It would now be of interest to mix different crystal shapes in the model and to introduce a grain size distribution instead of monodispersed particles. According to our results, we expect to deduce snow grain size information from bidirectional reflectance but not from polarized reflectances. Polarization provides additional information which could help to discriminate between clouds and snow, at least for water clouds over snow cover, and to characterize the aerosols.

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References

- Bohren, C. F., Multiple scattering of light and some of its observable consequences, *Am. J. Phys.*, 55, 524-533, 1987.
- Bourdelle, B., and M. Fily, Snow grain size determination from Landsat imagery over Terre Adélie, Antarctica, *Ann. Glaciol.*, 17, 86-92, 1993.
- Bréon, F.M., D. Tanré, P. Lecomte, and M. Herman, Polarized reflectance of bare soils and vegetation: measurements and model, *IEEE Trans. Geosci. Remote Sens.*, 33(2), 487-499, 1995.
- Brognez, G., Light scattering by finite hexagonal crystals arbitrarily oriented in space, paper presented at the International Radiation Symposium, Lille, France, 18-24 August, 64-67, A. Deepak Publishers, 1988.
- Brun, E., and E. Pahaut, An efficient method for a delayed and accurate characterization of snow grains from natural snowpacks, *J. Glaciol.*, 37(127), 420-422, 1991.
- Cai, Q., and K.N. Liou, Polarized light scattering by hexagonal ice crystals: Theory, *Appl. Opt.*, 21(19), 3569-3580, 1982.
- Chandrasekhar, S., *Radiative Transfer*, 393 pp, Dover. Press, New York, 1960.
- Coleman, R.F., and K.N. Liou, Light scattering by hexagonal ice crystals, *J. Atmos. Sci.*, 38, 1260-1270, 1981.
- Colin, L., The Pioneer Venus program, *J. Geophys. Res.*, 85, 7575-7598, 1980.
- De Haan, J.F., P.B. Bosma, and J.W. Hovenier, The adding method for multiple scattering calculations of polarized light, *Astron. Astrophys.*, 183, 371-397, 1987.
- Deschamps, P.Y., F.M. Bréon, M. Leroy, A. Podaire, A. Bricaud, J.C. Buriez, and G. Sèze, The POLDER mission: Instrument characteristics and scientific objectives, *IEEE Trans. Geosci. Remote Sens.*, 32, 598-615, 1994.
- Deuzé, J.L., M. Herman, and R. Santer, Fourier series expansion of the transfer equation in the atmosphere-ocean system, *J. Quant. Spectr. Radiat. Transfer*, 41(6), 483-494, 1989.
- Dollfus, A., Polarization studies of planets, in *Planets and Satellites*, edited by G. Kuiper and B. Middlehurst, Chicago Univ. Press, Chicago, Ill., 1961.
- Dozier, J., Spectral signature of Alpine snow cover from the Landsat Thematic Mapper, *Remote Sens. Environ.*, 28, 9-22, 1989.
- Fily, M., B. Bourdelle, J.P. Dedieu, and C. Sergeant, Comparisons of in situ and Landsat Thematic Mapper derived snow grain characteristics in the Alps, *Remote Sens. Environ.*, 59, 452-460, 1997.
- Fily, M., C. Leroux, J. Lenoble, and C. Sergeant, Terrestrial snow studies from remote sensing in the solar spectrum and in the thermal infrared, *Solar System Ices*, edited B. Schmitt, pp. 421-441, Astrophys. and Space Sci. Library, Kluwer Acad., 1998.
- Goloub, P., M. Herman, J.L. Deuzé, and R. Frouin, Contrast between polarization properties of snow/ice and clouds, *Antarc. J. US*, 27(5), 199-201, 1992.
- Grenfell, T.C., A radiative transfer model for sea ice with vertical structure variations, *J. Geophys. Res.*, 96, 16,991-17,001, 1992.

- Grenfell, T.C., and G.A. Maykut, The optical properties of ice and snow in the Arctic Basin, *J. Glaciol.*, 18, 445-463, 1977.
- Grenfell, T.C., S.G. Warren, and P.C. Mullen, Reflection of solar radiation by the Antarctic snow surface at ultraviolet, visible and near-infrared wavelengths, *J. Geophys. Res.*, 99, 18, 669-18,684, 1994.
- Hall, D.K., J.L. Foster, and A.T.C. Chang, Reflectance of snow as measured in situ and from space in sub-arctic areas in Canada and Alaska, *IEEE Trans. Geosci. Remote Sens.*, 30(3), 634-637, 1992.
- Hansen, J.E., and J.W. Hovenier, Interpretation of the polarization of Venus, *J. Atmos. Sci.*, 31, 1137-1160, 1974.
- Hapke, B., *Theory of Reflectance and Emittance Spectroscopy*, 455 pp, Cambridge Univ. Press, New York., 1993.
- Hovenier, J.W., and C.V.M. Van der Mee, Fundamental relationships relevant to the transfer of polarized light in a scattering atmosphere, *Astron. Astrophys.*, 128, 1-16, 1983.
- Iaquinta, J., H. Isaka, and P. Personne, Scattering phase function of bullet rosette ice crystals, *J. Atmos. Sci.*, 52(9), 1401-1413, 1994.
- Leroux, C., J. Lenoble, G. Brogniez, J.W. Hovenier, and J.F. De Haan, A model for the bidirectional polarized reflectance of snow, in press *J. Quant. Spectr. Radiat. Transfer*, 1998.
- Lesaffre, B., E. Pougatch, and E. Martin, Objective determination of snow grain characteristics from images, *Ann. Glaciol.*, in press, 1998.
- Marshall, S., and R.J. Oglesby, An improved snow hydrology for GCMs, 1, Snow cover fraction, albedo, grain size, and age, *Clim. Dyn.*, 10, 21-37, 1994.
- Sergeant, C., E. Pougatch, M. Sudul, and B. Bourdelles, Experimental investigation of optical properties for various types of snow, *Ann. Glaciol.*, 17, 281-287, 1993.
- Sergeant, C., C. Leroux, E. Pougatch, and F. Guirado, Hemispherical-directional reflectance measurements of natural snows in the 0.9-1.45 μm range, Comparison with the adding-doubling modelling, *Ann. Glaciol.*, in press, 1998.
- Steffen, K., Bidirectional reflectance of snow at 500-600 nm, Large scale effects of seasonal snow cover edited by Goodison, *IAHS 166*, pp. 415-425, 1987.
- Steffen, K., W. Abdalati, and J. Stroeve, Climate sensitivity studies of the Greenland ice sheet using satellite AVHRR, SMMR, SMM/I, and in situ data, *Meteorol. Atmos. Phys.*, 51, 239-258, 1993.
- Takano, Y., and K. Jayawera, Scattering phase matrix for hexagonal ice crystals computed from ray optics, *Appl. Opt.*, 24(19), 3254-3263, 1985.
- Takano, Y., and K.N. Liou, Radiative transfer in cirrus clouds, III, Light scattering by irregular ice crystals, *J. Atmos. Sci.*, 52(7), 818, 1995.
- Van de Hulst, H.C., *Light Scattering by Small Particles*, 470 pp, Dover Publications, New York 1981.
- Vermote, E.F., D. Tanré, J.L. Deuzé, M. Herman, and M. Morcrette, Second simulation of the satellite signal in the solar spectrum: An overview, *IEEE Trans. Geosci. Remote Sens.*, 35(3), 675-686, 1997.
- Warren, S.G., Optical properties of snow, *Rev. Geophys. Space Phys.*, 20(1), 66-89, 1982.
- Warren, S.G., Optical constant of ice from the ultraviolet to the microwave snow, *Appl. Opt.*, 23(1), 66-89, 1984.
- Wendling, P., R. Wendling, and H.K. Weickman, Scattering of solar radiation by hexagonal ice crystals, *Appl. Opt.*, 18, 2663-2671, 1979.
- Wiscombe, W.J., and S.G. Warren, A model for the spectral albedo of snow, I, Pure snow, *J. Atmos. Sci.*, 37, 2712-2733, 1980.
- phenomenon?, *Science*, 153, 56-60, 1966.
- Ottermann, J., and G.H. Weiss, Reflection from a field of randomly located vertical protrusions, *Appl. Opt.*, 23(12), 1931-1936, 1984.
- Parish, T.R., and D.H. Bromwich, The surface windfield over the Antarctic ice sheets, *Nature*, 328, 51-54, 1987.
- Roujean, J.L., M. Leroy, and P.Y. Deschamps, A bidirectional reflectance model of Earth's surface for correction of remote sensing data, *J. Geophys. Res.*, 97(D18), 20,455-20,468, 1992.
- Vermote, E. F., D. Tanré, J.L. Deuzé, M. Herman, and J.J. Morcrette, Second Simulation of the Satellite Signal in the Solar Spectrum, 6S: An overview, *IEEE Trans. Geosci. Remote Sens.*, 35(3), 675-686, 1997.
- Warren, S.G., and A.D. Clarke, Soot in the atmosphere and snow surface of Antarctica, *J. Geophys. Res.*, 95, 1811-1816, 1990.
- Wendler, G., and J. Kelley, On the albedo of snow in Antarctica: A contribution to I.A.G.O., *J. Glaciol.*, 34(116), 19-25, 1988.
- Wiscombe, W.J., and S.G. Warren, A model for the spectral albedo of snow I: Pure snow, *J. Atmos. Sci.*, 37, 2712-2733, 1980.

J.-L. Deuzé, P. Goloub, Laboratoire d'Optique Atmosphérique, Université des Sciences et Techniques de Lille, 59655 Villeneuve d'Ascq, France.

M. Fily and C. Leroux, Laboratoire de Glaciologie et Géophysique de l'Environnement, CNRS/UJF, 54 rue Molière, BP 96, 38402 St Martin d'Hères Cedex, France.

C. Sergeant, Centre National de Recherches Météorologiques, Centre d'Etudes de la Neige, Météo France, 1441 rue de la Piscine, 38406 Saint Martin d'Hères Cedex, France.

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